

A "quantum" Ramsey theorem for operator systems

Nik Weaver — Department of Mathematics, Washington University, Saint Louis, MO 63130

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ABSTRACT

Let $\mathcal{V}$ be a linear subspace of $M_n(\mathbb{C})$ which contains the identity matrix and is stable under the formation of Hermitian adjoints. We prove that if n is sufficiently large then there exists a rank k orthogonal projection P such that $\dim(P\mathcal{V}P) = 1$ or k^2 .

Posed as Severini's question — a quantum analogue of the classical Ramsey theorem. An operator system in finite dimensions is a unital, $*$ -closed linear subspace $\mathcal{V} \subseteq M_n(\mathbb{C})$. In quantum error correction, operator systems play the role classically played by confusability graphs.

3

THEOREMS

7

LEMMAS

3

PROPOSITIONS

3

DEFINITIONS

Definition 1.1 Quantum k -clique / quantum k -anticlique

§1 · p.2

Let $\mathcal{V} \subseteq M_n$ be an operator system. A **quantum k -clique** of $\mathcal{V}$ is an orthogonal projection $P \in M_n$ (i.e., $P = P^2 = P^*$) whose rank is k , such that $P\mathcal{V}P = \{PAP : A \in \mathcal{V}\}$ is maximal; that is, $P\mathcal{V}P = PM_nP \cong M_k$, or equivalently, $\dim(P\mathcal{V}P) = k^2$. A **quantum k -anticlique** of $\mathcal{V}$ is a rank k projection P such that $P\mathcal{V}P$ is minimal; that is, $P\mathcal{V}P = \mathbb{C} \cdot P \cong M_1$, or equivalently, $\dim(P\mathcal{V}P) = 1$.

NOTE

The anticlique definition matches the quantum-error-correction notion of a code: the range of a projection P with $P\mathcal{V}P = \mathbb{C} \cdot P$ [3].

Proposition 2.1 Diagonal system has no quantum anticlique ($k \geq 2$)

§2 · p.3

D_n has no quantum k -anticlique for $k \geq 2$.

PROOF SKETCH

If P has rank $k \geq 2$, then since each $\text{rank}(E_{ii}) = 1$, each $\text{rank}(PE_{ii}P) \in \{0, 1\}$. They cannot all be zero (else $P = \sum PE_{ii}P = 0$). Pick i with rank-1 $PE_{ii}P$; it cannot lie in $\mathbb{C} \cdot P$ (which has rank 0 or k), so $PD_nP \neq \mathbb{C} \cdot P$.

Proposition 2.2 Diagonal system has a quantum k -clique when n is large

§2 · p.3

If $n \geq k^2 + k - 1$ then D_n has a quantum k -clique.

PROOF SKETCH

Take $n = k^2 + k - 1$. Pick k^2 vectors $v_1, \dots, v_{k^2} \in \mathbb{C}^k$ whose rank-1 outer products $v_i v_i^*$ span M_k (e.g. $e_i, e_i + e_j, e_i + i e_j$). Identify $\mathbb{C}^n \cong \mathbb{C}^k \oplus \mathbb{C}^{k^2-1}$ and extend each v_i to a mutually orthogonal $w_i \in \mathbb{C}^n$ by appending a triangular block. Let P project onto $\mathbb{C}^k$; then PD_nP (relative to the basis $w_i / \|w_i\|$) contains $v_i v_i^*$, so $\dim(PD_nP) = k^2$.

NOTE

Improved later in Lemma 4.3. Note $n \geq k^2$ is necessary since $\dim(D_n) = n$ but $\dim(PD_nP) = k^2$ is required.

Proposition 2.3 Arbitrarily large systems may lack quantum 3-cliques

§2 · p.3

Let $\mathcal{V}_n = \text{span}\{I_n, E_{11}, E_{12}, \dots, E_{1n}, E_{21}, \dots, E_{n1}\} \subseteq M_n$. Then $\mathcal{V}_n$ has no quantum 3-cliques.

PROOF SKETCH

For any projection P : if $Pe_1 = 0$ then $PE_{1i}P = PE_{i1}P = 0$ and P is an anticlique. Otherwise, in an orthonormal basis $f_1 = Pe_1 / \|Pe_1\|, f_2, \dots, f_k$ of $\text{ran}(P)$, the compression $P\mathcal{V}_nP$ is exactly $\mathcal{V}_k \subseteq M_k$, of dimension $2k$. For $k \geq 3$, $2k < k^2$, so P is not a clique.

NOTE

Conjectured by M. Jury.

Lemma 3.1 Low-dimensional orthocomplement is spanned by rank-2 Hermitians

§3 · p.4

Let $\mathcal{V} \subseteq M_n$ be an operator system and suppose $\dim(\mathcal{V}) \leq 3$. Then its Hilbert-Schmidt orthocomplement is spanned by rank-2 Hermitian matrices.

PROOF SKETCH

Work in M_n^h . Suppose $B \in \mathcal{V}_0^\perp$ is Hermitian, nonzero, and orthogonal to the real span $\mathcal{W}_0$ of rank-2 Hermitians in $\mathcal{V}_0^\perp$. Decompose $B = B_+ - B_-$ (positive/negative parts), $\alpha = \text{Tr}(B_+) = \text{Tr}(B_-)$.

By convexity of the joint numerical range of three Hermitian matrices [1], pick unit vectors v, w with $\langle Av, v \rangle = \frac{1}{\alpha} \text{Tr}(AB_+)$ and $\langle Aw, w \rangle = \frac{1}{\alpha} \text{Tr}(AB_-)$ for $A \in \{A_1, A_2, B\}$. Then $C = \alpha(vv^* - ww^*)$ is rank-2 Hermitian, lies in $\mathcal{W}_0$, and is not orthogonal to B — contradiction.

Lemma 3.2 Four-dimensional $\mathcal{V} \subseteq M_3$ has a quantum 2-clique

§3 · p.4

Let $\mathcal{V} \subseteq M_3$ be an operator system and suppose $\dim(\mathcal{V}) = 4$. Then $\mathcal{V}$ has a quantum 2-clique.

PROOF SKETCH

Write $\mathcal{V} = \text{span}\{I_3, A_1, A_2, A_3\}$ with A_i Hermitian. Diagonalize A_1 so it has the form $\text{diag}(0, 1, a)$, and normalize A_2, A_3 so their (i, i) -entries vanish for $i \leq 2$. With $v = (1, \alpha, 0)^T$, $w = (1, 0, \beta)^T$, the 4×4 matrix of inner products $\langle A_i v, v \rangle, \langle A_i v, w \rangle, \langle A_i w, v \rangle, \langle A_i w, w \rangle$ has a nonzero determinant (polynomial in α, β) unless one of A_2, A_3 is a scalar multiple of the other.

Theorem 3.3 Quantum 2-cliques exist when $\dim(\mathcal{V}) \geq 4$

§3 · p.5

Let $\mathcal{V} \subseteq M_n$ be an operator system and suppose $\dim(\mathcal{V}) \geq 4$. Then $\mathcal{V}$ has a quantum 2-clique.

PROOF SKETCH

WLOG $\dim(\mathcal{V}) = 4$, $\mathcal{V} = \text{span}\{I_n, A_1, A_2, A_3\}$. Find a rank- ≤ 3 projection P so that PI_nP, PA_1P, PA_2P are linearly independent (split on whether A_1, A_2 are jointly diagonalizable). Pick $B \in M_n$ making the compression 4-dimensional and apply Lemma 3.2 to get a rank-2 $Q \leq P$ with $\{QI_nQ, QA_1Q, QA_2Q, QBQ\}$ independent. If $\{QI_nQ, QA_1Q, QA_2Q, QA_3Q\}$ is also independent, done; otherwise use Lemma 3.1 to find a rank-2 Hermitian $C = vv^* - ww^*$ detecting A_3 ; extend Q by v (or w) to a rank-3 Q' on which the four compressions are independent, then re-apply Lemma 3.2.

NOTE

Sharp: the result fails for arbitrary 4-dim subspaces, e.g. $\text{span}\{E_{11}, E_{12}, E_{13}, E_{14}\} \subset M_4$ has no rank-2 P with $\dim(PVP) > 2$.

Lemma 4.1 Diagonal-contained systems: dimension thresholds for cliques/anticliques

§4 · p.6

Suppose the operator system $\mathcal{V}$ is contained in D_n . If $\dim(\mathcal{V}) \geq k^2 + k - 1$ then $\mathcal{V}$ has a quantum k -clique. If $\dim(\mathcal{V}) \leq \frac{n-k}{k-1}$ then $\mathcal{V}$ has a quantum k -anticlique. If $n \geq k^3 - k + 1$ then $\mathcal{V}$ has either a quantum k -clique or a quantum k -anticlique.

PROOF SKETCH

Clique case: choose $m = k^2 + k - 1$ indices on which $\mathcal{V}$ compresses faithfully, then invoke Prop 2.2.
 Anticlique case: Tverberg's generalization of Radon's theorem [6, 7], packaged as in Theorem 4 of [4].
 Combining the two cases gives the cubic-in- k threshold via a little algebra.

Lemma 4.2 Gramian-extension to make vectors orthonormal-of-equal-norm

§4 · p.6

Let $v_1, \dots, v_r \in \mathbb{C}^s$. Then there are vectors $w_1, \dots, w_r \in \mathbb{C}^{r-1}$ such that the vectors $v_i \oplus w_i \in \mathbb{C}^{s+r-1}$ are pairwise orthogonal and all have the same norm.

PROOF SKETCH

Let G be the Gramian of (v_i) . Since $\text{rank}(\|G\|I_r - G) \leq r - 1$, one can realize $\|G\|I_r - G$ as the Gramian of some $w_i \in \mathbb{C}^{r-1}$. The Gramian of $(v_i \oplus w_i)$ is then $\|G\|I_r$.

NOTE

Improved by M. Jury.

Lemma 4.3

Quantum k -clique from upper-triangular-supported Hermitians (improves Prop 2.2)

§4 · p.6

Let $n = k^2 + k - 1$ and suppose $A_1, \dots, A_{k^2}$ are Hermitian matrices in M_n such that for each i , $\langle A_i e_i, e_i \rangle = 1$, and $\langle A_i e_r, e_s \rangle = 0$ whenever $\max\{r, s\} > i$. Then $\mathcal{V} = \text{span}\{I, A_1, \dots, A_{k^2}\}$ has a quantum k -clique.

PROOF SKETCH

Inductively pick $v_1, \dots, v_{k^2} \in \mathbb{C}^k$ so the compressions $A'_i = \sum_{r,s \leq i} a_{rs}^i v_r v_s^*$ are linearly independent. The freedom $\tilde{u} = u + v_i$ in writing $A'_i = B' + \tilde{u}\tilde{u}^*$ suffices because rank-1 updates span M_k . Then use Lemma 4.2 to assemble an orthonormal-of-equal-norm extension $f_i = (v_i \oplus w_i)/N$, yielding the desired clique.

Lemma 4.4 Quantum k -clique from off-diagonal-supported matrices (technical)

§4 · p.7

Let $n = k^4 + k^3 + k - 1$ and $\mathcal{V} \subseteq M_n$ an operator system containing matrices $A_1, \dots, A_{k^4+k^3}$ with $\langle A_i e_i, e_{i+1} \rangle \neq 0$ and $\langle A_i e_r, e_s \rangle = 0$ when $\max\{r, s\} > i + 1$ and $r \neq s$. Then $\mathcal{V}$ has a quantum k -clique.

PROOF SKETCH

Either some block of $k^2 + k - 1$ matrices has linearly independent r -tails — then Lemma 4.1 gives a diagonal-style k -clique — or one can construct orthonormal $v_1, \dots, v_{k^2}$ and Hermitian

$B_1, \dots, B_{k^2} \in \mathcal{V}$ whose compressions to $\text{span}\{v_1, \dots, v_{k^2}, e_{k^4+k^3+1}, \dots, e_{k^4+k^3+k-1}\}$ satisfy the hypotheses of Lemma 4.3.

Theorem 4.5 **MAIN THEOREM — quantum Ramsey for operator systems**

§4 · p.7

Every operator system in $\mathcal{M}_{8k^{11}}$ has either a quantum k -clique or a quantum k -anticlique.

PROOF SKETCH

Set $n = 8k^{11}$. Iteratively pick unit vectors $v_1, v_2, \dots$ so that $\dim((\mathcal{V}v_1)^\perp \cap \dots \cap (\mathcal{V}v_{r-1})^\perp \cap (\mathcal{V}v_r)) < 8k^8$. If the construction lasts k^3 steps, the compression to $\text{span}\{v_1, \dots, v_{k^3}\} \cong \mathcal{M}_{k^3}$ sits inside $\mathcal{D}_{k^3}$, so Lemma 4.1 supplies a clique or anticlique. Otherwise, in the residual subspace F , build a chain $w_1, w_2, \dots, w_r$ and matrices $A_1, \dots, A_{r-1} \in \mathcal{V}'$ with $w_{r+1} = A_r w_r$ orthogonal to $(\text{span}\{w_j, A_i w_j, A_i^* w_j : i \leq r-1, j \leq r\})^\perp$.

NOTE

The constants can be improved only marginally. Determining optimal bounds on quantum Ramsey numbers is likely open-ended, as in the classical case.

Definition 5.1 **Quantum graph on $\mathcal{M}$ (from [8, Def 2.6(d)])**

§5 · p.8

Let $\mathcal{M}$ be a unital $*$ -subalgebra of $\mathcal{M}_n$. A **quantum graph on $\mathcal{M}$** is an operator system $\mathcal{V} \subseteq \mathcal{M}_n$ which satisfies $\mathcal{M}'\mathcal{V}\mathcal{M}' = \mathcal{V}$.

NOTE

Representation-independent: $*$ -isomorphic $\mathcal{M}, \mathcal{N}$ give a natural correspondence of quantum graphs ([8, Thm 2.7]).

Definition 5.2 **Quantum k -clique / k -anticlique relative to $\mathcal{M}$**

§5 · p.9

Let $\mathcal{M} \subseteq \mathcal{M}_n$ be a unital $*$ -subalgebra and $\mathcal{V} \subseteq \mathcal{M}_n$ a quantum graph on $\mathcal{M}$. A rank k projection $P \in \mathcal{M}$ is a **quantum k -clique** if $P\mathcal{V}P = P\mathcal{M}_n P$, and a **quantum k -anticlique** if $P\mathcal{V}P = P\mathcal{M}'P$.

NOTE

Crucially, P must belong to $\mathcal{M}$. When $\mathcal{M} = \mathcal{M}_n$, this reduces to Definition 1.1; when $\mathcal{M} = \mathcal{D}_n$, it reproduces the classical k -clique/ k -anticlique notions.

Lemma 5.3 **Quantum Ramsey for quantum graphs of the form $\mathcal{M}_n \otimes I_d$**

§5 · p.9

Let $\mathcal{V} \subseteq M_{nd} \cong M_n \otimes M_d$ be a quantum graph on $M_n \otimes I_d$. If $nd \geq 8k^{11}$ then there is a projection in $M_n \otimes I_d$ whose rank is at least k and which is either a quantum clique or a quantum anticlique of $\mathcal{V}$.

PROOF SKETCH

$\mathcal{V} = \mathcal{W} \otimes M_d$ for some operator system $\mathcal{W} \subseteq M_n$. If $d = 1$, apply Thm 4.5. If $d \geq k$, any $P \otimes I_d$ with rank-1 P works. For $2 \leq d < k$: algebra shows $n \geq 8\lceil k/d \rceil^{11}$, so Thm 4.5 gives a quantum $\lceil k/d \rceil$ -clique/anticlique $Q \in M_n$, and $Q \otimes I_d$ is the required object.

NOTE

Can only promise rank $\geq k$, since ranks in $M_n \otimes I_d$ are multiples of d .

Theorem 5.4 Common generalization of classical and quantum Ramsey theorems

§5 · p.10

For every k there exists n such that if $\mathcal{M}$ is a unital $*$ -subalgebra of M_n and $\mathcal{V} \subseteq M_n$ is an operator system satisfying $\mathcal{M}'\mathcal{V}\mathcal{M}' = \mathcal{V}$, then there is a projection $P \in \mathcal{M}$ whose rank is at least k and such that $PVP = PM_nP$ or $PM'P$.

PROOF SKETCH

Take $n = 8k^{11} \cdot R(k, k)$ where $R(k, k)$ is the classical Ramsey number. Write $\mathcal{M} = \bigoplus_{i=1}^r (M_{n_i} \otimes I_{d_i})$. If $r \leq R(k, k)$, some $n_i d_i \geq 8k^{11}$ and Lemma 5.3 applies to that block. If $r > R(k, k)$, pick rank-1 $Q_i \in M_{n_i}$, let $Q = \bigoplus Q_i \otimes I_{d_i}$; then $Q\mathcal{V}Q$ corresponds to a simple graph G on $\{1, \dots, r\}$ by [8, Thm 2.7]; the classical Ramsey theorem applied to G yields the required diagonal projection.

NOTE

Recovers Thm 4.5 when $\mathcal{M} = M_n$, and the classical Ramsey theorem when $\mathcal{M} = D_n$ (both with worse constants). The gap between classical (exponential growth) and quantum (polynomial growth) Ramsey numbers is explained in the discussion: the classical bound's exponential dependence comes from needing $\mathcal{W}\mathcal{V}$ of small codimension rather than large dimension.

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